

Final Problem Set

Cerebellar Expansion · Attractor Networks · Learning Systems

Neurobiological Basis of Intelligence

Total: 100 pts

Problem 1 — Cerebellar Expansion Recoding

[35 pts]

In the cerebellar cortex, mossy fibers carry input signals to granule cells, which then project via parallel fibers to Purkinje cells. Each granule cell receives K mossy fiber inputs (in mammals, $K \approx 4$) and acts as a coincidence detector: it fires only when all K of its inputs are simultaneously active.

Part A: Decorrelation by expansion

Two mossy fiber input patterns each activate L fibers out of a total population of M mossy fibers. The two patterns share W active fibers in common, so the overlap at the mossy fiber level is W/L .

Each granule cell receives exactly K mossy fiber inputs and fires only if all K are active.

- Show that the number of granule cells activated by a single pattern (with L active fibers) is at most $\binom{L}{K}$, and the number activated by both patterns simultaneously is at most $\binom{W}{K}$.
- Derive the overlap in the granule cell representation:

$$\text{Overlap}_{\text{GC}} = \frac{\binom{W}{K}}{\binom{L}{K}} = \frac{W(W-1)\cdots(W-K+1)}{L(L-1)\cdots(L-K+1)}$$

and show that for $L, W \gg K$, this approaches $(W/L)^K$.

- For $L = 100$, $W = 40$ (40% overlap at the mossy fiber level), compute the exact overlap in the granule cell representation for $K = 2, 3, 4, 5, 7$. Compare each to the approximation $(W/L)^K$. Comment on how rapidly the overlap decreases and the quality of the approximation.
- Explain intuitively why the overlap decreases exponentially with K . Use an analogy: if two people each know 100 colleagues and share 40 in common, what is the probability they share the same specific group of K colleagues?

Part B: Optimal number of inputs per granule cell

Suppose the cerebellum has M mossy fibers in total, and each granule cell randomly and independently chooses K of the M mossy fibers as its inputs. There are N granule cells.

- The total number of distinct input combinations is $\binom{M}{K}$. If each granule cell independently draws one of these $\binom{M}{K}$ combinations at random, what is the probability p that all N granule cells receive *different* combinations? Express your answer as a product.

Hint: This is analogous to the birthday problem. The first granule cell can choose any combination; the second must avoid the first's choice; the third must avoid the first two; and so on.

- (b) Show that p can be written as:

$$p = \prod_{i=0}^{N-1} \left(1 - \frac{i}{\binom{M}{K}} \right)$$

For $N \ll \binom{M}{K}$, use the approximation $\ln(1-x) \approx -x$ to show that:

$$p \approx \exp\left(-\frac{N(N-1)}{2\binom{M}{K}}\right)$$

- (c) For fixed N and M , p depends on K through $\binom{M}{K}$. As K increases from 1, $\binom{M}{K}$ first increases (more distinct combinations), reaches a maximum at $K = M/2$, then decreases. Thus p also has a maximum as a function of K .

Using $N = 21,000$ and $M = 7,000$, numerically compute and plot p as a function of K for $K = 1, 2, \dots, 20$. Find the value of K at which p first exceeds 95% of its maximum value. Compare this optimal K to the actual value $K \approx 4$ observed in the mammalian cerebellum. Comment on the result.

- (d) The arguments in Part A and Part B both point to $K \approx 4$ as a sweet spot, but for different reasons. Part A is about *decorrelation* (separating overlapping inputs). Part B is about *diversity* (ensuring each granule cell is unique). Are these two objectives complementary or in tension? Could the brain be optimizing for both simultaneously?

Part C: Connection to machine learning

- (a) The granule cell layer implements a nonlinear feature expansion $\mathbf{x} \mapsto \boldsymbol{\phi}(\mathbf{x})$, followed by a linear readout at the Purkinje cell: $y = \mathbf{w}^\top \boldsymbol{\phi}(\mathbf{x})$. The weights $\mathbf{w}$ are learned via the LTD rule $\Delta w_i = -\eta \cdot \phi_i \cdot e_{\text{CF}}$. Identify the correspondence between this architecture and:
- A single-layer perceptron with random features
 - Kernel regression with random Fourier features
 - A reservoir computer / echo state network
- (b) The cerebellum has only *one* layer of adjustable weights (parallel fiber $\rightarrow$ Purkinje cell). How does the expansion recoding compensate for the lack of depth? Under what conditions is a single linear layer on top of random nonlinear features as powerful as a deep network? What are the limitations?

Problem 2 — Hopfield Networks and Grid Cell Attractors

[35 pts]

Both Hopfield networks and grid cell attractor models use recurrent neural networks whose dynamics converge to stable states. But the nature of these attractors is fundamentally different.

Part A: Hopfield network capacity

Consider a Hopfield network with N neurons storing P random binary patterns $\xi^1, \dots, \xi^P \in \{-1, +1\}^N$ using the outer product rule:

$$W_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^\mu \xi_j^\mu, \quad W_{ii} = 0.$$

- (a) Suppose the network state is $\mathbf{x} = \xi^1$ (one of the stored patterns). Show that the local field at neuron i is:

$$h_i = \sum_j W_{ij} \xi_j^1 = \xi_i^1 + \frac{1}{N} \sum_{\nu=2}^P \xi_i^\nu \left(\sum_j \xi_j^\nu \xi_j^1 \right)$$

Identify the signal term and the cross-talk noise term. Assuming the patterns are random and i.i.d., argue that the noise term has mean 0 and variance $(P-1)/N$.

- (b) Using the result from (a), explain why the pattern ξ^1 is a stable fixed point when P/N is small, and why retrieval fails catastrophically when P/N exceeds $\alpha_c \approx 0.138$. Why is the transition sharp rather than gradual?
- (c) The modern Hopfield network (Ramsauer et al., 2020) uses the energy:

$$E = -\log \sum_{\mu=1}^P \exp(\xi^\mu \cdot \mathbf{x})$$

Write down the update rule $\mathbf{x}_{\text{new}} = \sum_{\mu} \text{softmax}(\beta \xi^\mu \cdot \mathbf{x}) \xi^\mu$ and explain why this leads to exponential storage capacity $P_{\text{max}} \sim \exp(\alpha N)$ instead of $P_{\text{max}} \sim N$. What is the connection to transformer self-attention?

Part B: Continuous attractors and grid cells

In the Burak & Fiete (2009) model, N^2 neurons are arranged on a 2D sheet with center-surround connectivity:

$$W_0(\mathbf{x}) = A_e e^{-|\mathbf{x}|^2/2\sigma_e^2} - A_i e^{-|\mathbf{x}|^2/2\sigma_i^2}, \quad \sigma_e < \sigma_i$$

- (a) Explain why this connectivity profile (local excitation, surround inhibition) produces a stable pattern of activity “bumps” via Turing instability. What determines the spacing of the bumps?
- (b) The full connectivity with the velocity shift mechanism is:

$$W_{ij} = W_0(\mathbf{x}_i - \mathbf{x}_j - l \hat{\mathbf{e}}_{\theta_j})$$

Explain how the shift $l \hat{e}_{\theta_j}$ breaks the symmetry of the weight matrix and converts a static bump into a velocity integrator. Why does the bump remain stationary when the velocity input is zero, despite the asymmetric weights?

- (c) The isotropic Gaussian $e^{-|\Delta \mathbf{r}|^2/2\sigma^2}$ is separable in orthogonal Cartesian coordinates:

$$e^{-(\Delta x^2 + \Delta y^2)/2\sigma^2} = e^{-\Delta x^2/2\sigma^2} \cdot e^{-\Delta y^2/2\sigma^2}$$

Explain why this separability means the 2D model on a *standard torus* (90° axes) can be decomposed into two *independent* 1D ring attractors, producing a square grid. Now consider oblique coordinates at 60°:

$$\Delta x^2 + \Delta y^2 = \Delta u^2 + \Delta u \Delta v + \Delta v^2$$

The cross-term $\Delta u \Delta v$ breaks separability. Explain why this means the two ring attractors on a *twisted torus* are *coupled*, and why the resulting grid pattern is hexagonal rather than square.

Part C: Comparing the two attractor systems

- (a) Draw or describe the energy landscape for a Hopfield network and for a grid cell continuous attractor network. In each case, what do the stable states look like? What happens when you perturb the state slightly away from a stable point?
- (b) The Hopfield network stores *discrete* memories at isolated energy minima. The grid cell network has a *continuous manifold* of equivalent states (all translations of the bump pattern). What are the computational advantages and disadvantages of each architecture?
- (c) Can a single network have both discrete and continuous attractor properties? Give an example from neuroscience where this might occur. *Hint:* Consider a network that must both integrate a continuous variable and remember discrete landmark locations.

Problem 3 — Comparing Learning Systems (Open-Ended)

[30 pts]

The brain uses multiple learning systems simultaneously, each implemented in a distinct anatomical circuit. In this problem, you will synthesize ideas across the full lecture series.

Part A: The eligibility trace across systems

The concept of an eligibility trace — a temporary biochemical tag at a synapse that marks it as “eligible” for modification by a later modulatory signal — appears in multiple brain systems with different timescales:

System	Timescale	Modulatory signal
Cortex (STDP)	~20–50 ms	Coincidence (pre/post)
Cerebellum (LTD)	~100–200 ms	Climbing fiber
Basal ganglia (three-factor)	~1–2 s	Dopamine

- (a) Why are the timescales different in each system? Relate the timescale to the temporal credit assignment problem each system faces. What would go wrong if the basal ganglia eligibility trace were as short as the cerebellar one? What would go wrong if the cerebellar trace were as long as the basal ganglia one?
- (b) In all three systems, the learning rule can be written in the general form:

$$\Delta w_{ij} = \eta \cdot \text{trace}(x_i, x_j) \cdot \text{modulator}$$

Identify the specific biological implementation of the “trace” and “modulator” in each system. What makes this general form biologically plausible (as opposed to backpropagation)?

Part B: A unified view

- (a) Fill in the following table with the biological implementation for each learning system:

	Error signal	Adjustable weights	Representation	Credit assignment
Cerebellum				
Basal ganglia				
Cortex (pred. coding)				
Hippocampus				

- (b) A student claims: “The cerebellum does supervised learning, the basal ganglia do reinforcement learning, the cortex does unsupervised learning, and the hippocampus does one-shot memory — these are four independent systems that don’t interact.” Critique this claim. Give at least three specific examples of interactions or dependencies between these systems.

- (c) Could an artificial neural network benefit from incorporating multiple specialized learning systems, analogous to the brain? Current large language models use a single learning algorithm (backpropagation through a transformer). What computational problems might be better solved by a Doya-style architecture with specialized modules for supervised, reinforcement, and unsupervised learning? What are the challenges in implementing such a system?